DEBT SUSTAINABILITY: THE ASSET PRICING APPROACH

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US DEBT EXCEEDS WHAT FUTURE SURPLUSES CAN SUPPORT

- Last time: the IGBC tells us $B_t = \mathbb{E}_t \sum M_{t,t+j} S_{t+j}$. But we left open the key question: what is the correct discount rate?
- Classic DSA uses r^f – but we showed that if surpluses are procyclical, $\text{cov}(M, S) < 0$ and discounting at r^f overstates fiscal capacity.
- Today: use tools from asset pricing to measure the correct discount rate and quantify fiscal capacity.
- The answer: the market value of US debt has consistently exceeded the risk-adjusted PDV of surpluses since the early 2000s.
- Main reference: Jiang, Lustig, Van Nieuwerburgh & Xiaolan (2023, *Annual Review of Financial Economics*).

ASSET PRICING REVIEW

AN ASSET IS A CLAIM ON FUTURE PAYMENTS

- The general framework dates back to Lucas (1978).
- An economy with $i = 1, \ldots, N$ assets. Each asset has a stream of real dividends $\left\{d_{i,t}\right\}_{t=0}^{\infty}$.
- A representative investor maximizes $\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$.
- Let $p_{i,t}$ be the price of asset i at time t (in units of goods).
- Question: what determines $p_{i,t}$?

OPTIMAL PORTFOLIO CHOICE YIELDS THE EULER EQUATION

- For each asset the investor holds, the optimality condition is:

$$p_{i,t} = \beta \, \mathbb{E}_t \frac{u'(c_{t+1})}{u'(c_t)} \left(p_{i,t+1} + d_{i,t+1} \right).$$

- This is the consumption-based asset pricing equation.
- Rearrange as

$$1 = \beta \, \mathbb{E}_t \frac{u'(c_{t+1})}{u'(c_t)} R_{i,t,t+1},$$

where $R_{i,t,t+1} := \frac{p_{i,t+1} + d_{i,t+1}}{p_{i,t}}$ is the gross return on asset i.

THE STOCHASTIC DISCOUNT FACTOR PRICES ALL ASSETS

- Define the stochastic discount factor (SDF): $M_{t,t+1} := \beta \frac{u'(c_{t+1})}{u'(c_t)}$.
- The asset pricing equation for every asset:

$$1 = \mathbb{E}_t M_{t,t+1} R_{i,t,t+1}.$$

- The risk-free rate $R^f_{t,t+1}$ satisfies

$$1 = R^f_{t,t+1} \mathbb{E}_t M_{t,t+1}.$$

- The SDF is one object that prices all assets simultaneously.

THE PRICE-DIVIDEND RATIO CAPTURES VALUATION

- Define the price-dividend ratio: $v_{i,t} := p_{i,t}/d_t$.
- The asset pricing equation becomes:

$$v_{i,t} = \mathbb{E}_t M_{t,t+1} \frac{d_{i,t+1}}{d_t} \left(1 + v_{i,t+1}\right).$$

- Useful when dividends grow over time: the price-dividend ratio is stationary even if prices and dividends trend upward.
- The debt-to-GDP ratio is the government's "price-dividend ratio."

EXPECTED RETURNS EQUAL THE RISK-FREE RATE PLUS A RISK PREMIUM

- Since $\mathbb{E}_t MR_i = \mathbb{E}_t M \cdot \mathbb{E}_t R_i + \text{cov}_t(M, R_i)$, we can write:

$$\mathbb{E}_t R_{i,t,t+1} = R^f_{t,t+1} - \frac{\text{cov}_t\left(M_{t,t+1}, R_{i,t,t+1}\right)}{\mathbb{E}_t M_{t,t+1}}.$$

- The risk premium depends on the covariance of the return with the SDF.
- Negative covariance (asset pays poorly in bad times) $\Rightarrow$ positive risk premium.
- Positive covariance (asset pays well in bad times) $\Rightarrow$ negative risk premium (insurance).

THE RISK PREMIUM REWARDS BEARING AGGREGATE RISK

- Example: $u(c) = \ln c$. Then $M_{t,t+1} = \beta \frac{c_t}{c_{t+1}}$.
- The SDF is high when consumption is low (recessions).
- An asset whose return is low when consumption is low has a negative covariance with the SDF.
- Investors demand a positive risk premium to hold this asset – it does not pay when they need it most.
- An asset that pays well in recessions provides insurance and earns a lower expected return.

RISK EXPOSURE AND PRICE OF RISK DETERMINE THE PREMIUM

- Decompose the risk premium:

$$\mathbb{E}_t R_{i,t,t+1} = R^f_{t,t+1} + \beta_{i,t}\lambda_t.$$

- Risk exposure (beta): $\beta_{i,t} := -\frac{\mathrm{cov}_t\left(M_{t,t+1}, R_{i,t,t+1}\right)}{\mathrm{var}_t M_{t,t+1}}$.
- Price of risk: $\lambda_t := \frac{\mathrm{var}_t M_{t,t+1}}{\mathbb{E}_t M_{t,t+1}}$.
- Risk exposure is asset-specific. Price of risk is common to all assets.
- The Sharpe ratio of an asset is its expected excess return per unit of volatility: $\xi_i = \frac{\mathbb{E}_t R_i - R^f}{\sigma(R_i)}$. An asset with $\beta_i = 1$ has Sharpe ratio $\xi = \lambda/\sigma$.
- Note: $\beta_{i,t}$ here is the asset's beta, not the discount factor β.

NO-ARBITRAGE IS ENOUGH FOR AN SDF TO EXIST

- So far we derived the SDF from the representative agent's problem.
- Fundamental theorem of asset pricing (Harrison & Kreps, 1979): (some) SDF exists under much weaker conditions – it is enough that there is no arbitrage.
- This is important: we do not need to take a stand on the specific model of preferences.
- We can use the SDF to price any cash flow stream – including government surpluses.
- This is the approach of Jiang et al. (2024, *Econometrica*).

VALUATION APPROACH TO DEBT SUSTAINABILITY

THE IGBC VALUES DEBT AS THE PDV OF FUTURE SURPLUSES

- Iterate the government budget constraint forward:

$$B_t = \mathbb{E}_t \sum_{j=1}^{T} M_{t,t+j} S_{t+j} + \mathbb{E}_t M_{t,t+T} B_{t+T},$$

  where $S_{t+j} = T_{t+j} - G_{t+j}$ is the primary surplus.
- Impose the transversality condition: $\lim_{T\to\infty} \mathbb{E}_t M_{t,t+T} B_{t+T} = 0$.
- The intertemporal government budget constraint (IGBC):

$$B_t = \mathbb{E}_t \sum_{j=1}^{\infty} M_{t,t+j} S_{t+j}.$$

THE IGBC IN LEVELS AND RATIOS TO GDP

- The IGBC as stated is in levels (dollars):

$$B_t = \mathbb{E}_t \sum_{j=1}^{\infty} M_{t,t+j} S_{t+j}.$$

- Dividing both sides by Y_t:

$$\frac{B_t}{Y_t} = \mathbb{E}_t \sum_{j=1}^{\infty} M_{t,t+j} \frac{Y_{t+j}}{Y_t} \frac{S_{t+j}}{Y_{t+j}}.$$

- We move between levels and ratios to GDP depending on context.

THE IGBC IS THE CAMPBELL-SHILLER DECOMPOSITION FOR GOVERNMENT DEBT

- A higher debt-to-output ratio today must be attributed to:
  1. Higher expected future primary surpluses (cash flows), or
  2. Lower expected future returns on debt (discount rates).
- Cochrane (2019): about half of the variation in debt/GDP is due to surpluses, half to discount rates.
- Jiang, Lustig, Van Nieuwerburgh & Xiaolan (2021): after correcting for small-sample bias, neither channel is statistically significant. Debt/GDP at t predicts debt/GDP at $t + T$.
- The market value of debt appears disconnected from fiscal fundamentals – foreshadowing the valuation puzzle we quantify below.

FISCAL CAPACITY: HOW MUCH WOULD YOU PAY FOR THE GOVERNMENT DEBT PORTFOLIO?

- Suppose an investor buys the entire stock of government debt and participates in all future issuances.
- The cash flow to this investor is $S_{t+j} = T_{t+j} - G_{t+j}$.
- Jiang et al. call the present value of these cash flows the fiscal capacity:

$$\text{Fiscal capacity} = \mathbb{E}_t \sum_{j=1}^{\infty} M_{t,t+j} S_{t+j}.$$

- The key question: what is the right discount rate for government surpluses?

RISK SHRINKS FISCAL CAPACITY BELOW THE RISK-NEUTRAL BENCHMARK

- Use the fact that $\mathbb{E}_t M_{t,t+j} S_{t+j} = \mathbb{E}_t M_{t,t+j} \cdot \mathbb{E}_t S_{t+j} + \text{cov}_t \left(M_{t,t+j}, S_{t+j} \right)$:

$$B_t = \sum_{j=1}^{\infty} \mathbb{E}_t M_{t,t+j} \cdot \mathbb{E}_t S_{t+j} + \sum_{j=1}^{\infty} \text{cov}_t \left(M_{t,t+j}, T_{t+j} \right) - \sum_{j=1}^{\infty} \text{cov}_t \left(M_{t,t+j}, G_{t+j} \right).$$

- Three components:
  1. PDV of surpluses at the risk-free rate.
  2. Covariance of SDF with tax revenue.
  3. Covariance of SDF with government spending.

IN A RISK-FREE WORLD, ONLY THE ABILITY TO GENERATE SURPLUSES MATTERS

- If there is no aggregate risk, the second and third terms vanish.
- Fiscal capacity is determined only by the level of future surpluses discounted at the risk-free rate.
- This is the standard approach in the macro literature on debt sustainability.
- But taxes and spending are correlated with the business cycle. What do the covariance terms imply?

PROCYCLICAL TAXES AND COUNTERCYCLICAL SPENDING REDUCE FISCAL CAPACITY

- Tax revenue is procyclical: high in expansions (M low), low in recessions (M high).
- $\Rightarrow \mathrm{cov}(M, T) < 0$: the tax covariance term lowers fiscal capacity.
- Government spending is countercyclical: transfers rise in recessions (M high).
- $\Rightarrow \mathrm{cov}(M, G) > 0$: the spending covariance term also lowers fiscal capacity.
- Both covariance terms work against the government.
- Fiscal capacity is most likely lower than the risk-neutral benchmark.

A STYLIZED MODEL ECONOMY

SETUP: CONSTANT TAX AND SPENDING RATIOS, RISKY GDP GROWTH

- Following Jiang et al. (2023, Section 3), consider a simple economy:
  - Average tax revenue to GDP: $\tau := \mathbb{E}[T/Y] = 25\%$.
  - Average government spending to GDP: $\gamma := \mathbb{E}[G/Y] = 22.5\%$.
  - Risk-free rate: $r^f = 1.5\%$.
  - Mean GDP growth rate: $x = 2\%$.
  - Volatility of GDP growth: $\sigma = 3\%$.
  - Initial GDP: $Y = \$10$ trillion.
- GDP growth is i.i.d. with risk. Tax and spending ratios are constant.
- All fiscal cash flows grow at the GDP rate, so we need to price a claim to GDP.

CALIBRATING THE GDP RISK PREMIUM

- Recall: the risk premium on any asset is $\beta_i \times \lambda$ (risk exposure $\times$ price of risk).
- In a one-factor model where GDP growth is the only shock with volatility $\sigma = 3\%$:
  - Normalize $\beta_Y = 1$ (GDP is the risk factor).
  - Then $\lambda = \xi\sigma$, where ξ is the Sharpe ratio on the GDP claim.
- The GDP risk premium is:

$$RP = \beta_Y \times \lambda = 1 \times \xi\sigma = 1 \times 1 \times 3\% = 3\%.$$

- Jiang et al. set $\xi = 1$: investors demand a Sharpe ratio of 1 on the GDP claim.
- Note: the equity risk premium ($\approx 7\%$) is higher because the stock market is a levered claim to GDP – fixed costs and corporate debt amplify output risk.
- The TVC holds as long as $\xi > 0.166$ (i.e. $RP > x - r^f = 0.5\%$), so $\xi = 1$ is well above the minimum.

VALUING A CLAIM TO GDP: THE GORDON GROWTH FORMULA

- Think of GDP as a "dividend" that grows at rate x on average. What is a claim to all future GDP worth?
- The present value of a growing perpetuity discounted at rate r:

$$PV = \frac{Y}{r - x} = Y \times \underbrace{\frac{1}{r - x}}_{pd \text{ ratio}}, \qquad \text{valid if } r > x.$$

- From the asset pricing framework: the correct discount rate for a risky cash flow includes a risk premium. GDP is risky, so $r_Y = r^f + RP = 1.5\% + 3\% = 4.5\%$. You will show this formally in a problem set.
- The formula works because $r^f + RP = 4.5\% > x = 2\%$, even though $r^f = 1.5\% < x = 2\%$.

THE CLAIM TO GDP IS WORTH $400 TRILLION

- Plug in the numbers:

$$r_Y = r^f + RP = 1.5\% + 3\% = 4.5\%$$

$$pd^Y = \frac{1}{r_Y - x} = \frac{1}{4.5\% - 2\%} = \frac{1}{2.5\%} = 40$$

$$PV(\text{GDP}) = Y \times pd^Y = \$10\text{T} \times 40 = \$400\text{T}$$

- The TVC is satisfied because $r^f + RP = 4.5\% > x = 2\%$.
- This holds even though $r^f = 1.5\% < x = 2\%$.

FISCAL CAPACITY EQUALS 100% OF GDP

- Start with $T = \tau Y$ and $G = \gamma Y$ (proportional to GDP), so $\beta_T = \beta_G = \beta_Y = 1$. Use the same *pd* ratio:

$$\frac{B}{Y} = \frac{\tau - \gamma}{r^f + RP - x} = \frac{25\% - 22.5\%}{1.5\% + 3\% - 2\%} = \frac{2.5\%}{2.5\%} = 1.$$

- Fiscal capacity = \$10T × 1 = \$10 trillion = 100% of GDP.
- Note: so far we have only corrected for GDP risk. We have not yet allowed taxes and spending to differ in their cyclicality – that will make things worse.

IGNORING THE RISK PREMIUM GIVES INFINITE FISCAL CAPACITY

- What if we discounted at the risk-free rate only ($RP = 0$)?

$$\frac{B}{Y} = \frac{\tau - \gamma}{r^f - x} = \frac{2.5\%}{1.5\% - 2\%} = \frac{2.5\%}{-0.5\%} < 0.$$

- The denominator is negative: $r^f < x$. The geometric sum diverges.
- Economically: each dollar of surplus grows faster than the discount rate, so the PDV is infinite.
- This is exactly the $r < g$ argument from last time (Blanchard 2019). It implicitly assumes the surplus is risk-free ($RP = 0$).
- Any $RP > x - r^f = 0.5\%$ makes the denominator positive and fiscal capacity finite. Our calibration uses $RP = 3\%$.

WHY LOW TREASURY YIELDS DO NOT MEAN DEBT IS CHEAP

- Common argument: “yields are low, so the government can borrow cheaply.”
- This conflates two objects:
  1. The Treasury yield – what bondholders earn. This reflects safety, liquidity, and Fed policy.
  2. The discount rate for surpluses – what the government must generate in risk-adjusted terms. This is $r^f + RP$.
- The Treasury yield can be low because of convenience yields or QE. But the surplus stream is still tied to GDP – its risk has not changed.
- As Jiang et al. (2022) put it: “Despite current low interest rates, we find that US fiscal capacity is more limited than commonly thought.”
- The Fed controls nominal short rates, not the GDP risk premium. Lowering yields does not lower the rate at which surpluses should be discounted.

THE TREASURY BALANCE SHEET: TAXES FUND SPENDING AND DEBT

- Present values of the tax and spending claims:

$$PV(T) = \tau \times Y \times 40 = 0.25 \times \$10 \times 40 = \$100 \text{ trillion}$$

$$PV(G) = \gamma \times Y \times 40 = 0.225 \times \$10 \times 40 = \$90 \text{ trillion}$$

Assets		**Liabilities**	
$PV(T)$	\$100	$PV(G)$	\$90
		D	\$10
Total	\$100	Total	\$100

THE TREASURY BALANCE SHEET IMPLIES A BETA FOR DEBT

- The balance sheet: $PV(T) = PV(G) + D$. Debt is a residual claim: taxes minus spending.
- β_T, β_G, β_D are the betas of each claim (same β_i as before): how much risk each cash flow carries. Recall $\beta_Y = 1$.
- By no-arbitrage, the risk of the assets must equal the risk of the liabilities:

$$\underbrace{PV(T)}_{100}\ \beta_T = \underbrace{PV(G)}_{90}\ \beta_G + \underbrace{D}_{10}\ \beta_D.$$

- Solving for β_D:

$$\beta_D = \frac{PV(T)}{D}\beta_T - \frac{PV(G)}{D}\beta_G.$$

- This is analogous to Modigliani-Miller: the beta of equity (debt) is determined by the beta of assets (taxes) and leverage.

GOVERNMENT DEBT INHERITS GDP RISK

- With $\beta_T = \beta_G = \beta_Y = 1$ (everything proportional to GDP):

$$\beta_D = \frac{100}{10} \cdot 1 - \frac{90}{10} \cdot 1 = 10 - 9 = 1.$$

- The leverage is enormous: $PV(T)/D = 10$, $PV(G)/D = 9$.
- Even though taxes and spending are equally risky, debt has $\beta_D = 1$.
- Intuition: investors who hold the debt portfolio are net long a claim to output. The output risk in spending does not fully offset the output risk in tax revenue.

INSURING TAXPAYERS MEANS LESS INSURANCE FOR BONDHOLDERS

- In reality, $\beta_T > \beta_Y > \beta_G$:
  - Tax revenue is more volatile than GDP (procyclical).
  - Government spending is less volatile than GDP (countercyclical transfers).
- The formula $\beta_D = \frac{PV(T)}{D}\beta_T - \frac{PV(G)}{D}\beta_G$ shows the trade-off. Note: $PV(T)$ and $PV(G)$ themselves change with betas, because each claim is discounted at $r_i = r^f + \beta_i\lambda$.
- If the government insures taxpayers (higher β_T, lower tax in recessions), there is less insurance for bondholders (higher β_D).
- This raises the risk premium on debt and lowers fiscal capacity.

RISK-FREE GOVERNMENT DEBT

MANUFACTURING RISK-FREE DEBT REQUIRES COUNTERCYCLICAL TAXES

- "Risk-free debt" means $\beta_D = 0$. From the beta formula:

$$\beta_T = \frac{PV(G)}{PV(G) + D} \beta_G < \beta_G.$$

- Since $D > 0$, the weight $\frac{PV(G)}{PV(G)+D} < 1$, so $\beta_T < \beta_G$: taxes must be safer than spending.
- This requires abandoning the proportional tax rule $T = \tau Y$ (which gives $\beta_T = 1$).
- Instead, tax revenue must be less sensitive to GDP than one-for-one: when Y falls, T falls by less, so the effective tax rate T/Y rises in recessions.
- We keep $\tau = \mathbb{E}[T/Y]$ as the average tax rate, but now $\beta_T < 1$ controls the cyclical response separately.
- This is counterfactual for the US, where tax revenue is strongly procyclical ($\beta_T > 1$).

FINDING β_T: PRESENT VALUES DEPEND ON BETAS

- The formula $\beta_T = \frac{PV(G)}{PV(G)+D}\beta_G$ looks simple, but $PV(T)$ and D depend on β_T through the discount rate:

$$r_T = r^f + \beta_T \cdot RP, \qquad \text{pd ratio on taxes} = \frac{1}{r_T - x}.$$

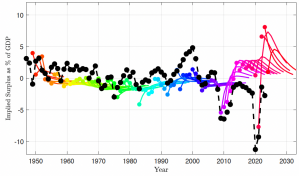
- Key simplification: $PV(G) = \gamma Y/(r_G - x) = 90$ does not depend on β_T (since $\beta_G = 1$ is unchanged).
- So there is one equation in one unknown ($PV(T)$):

$$PV(T) = \frac{\tau \cdot Y}{r^f + \frac{90}{PV(T)} \cdot RP - x} \implies PV(T) = 100.$$

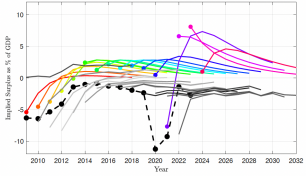
- Then: $D = 100 - 90 = 10$, $\beta_T = 90/100 = 0.9$, and $r_T = 1.5\% + 0.9 \times 3\% = 4.2\%$.

RISK-FREE DEBT IMPLIES SURPLUSES THAT NEVER MATERIALIZED

Figure 2: Expected Surpluses



(a) Full Time-Series



(b) Since the GFC

Expected surpluses implied by $\beta_D = 0$ vs. realized.

THE TREASURY BALANCE SHEET WITH RISK-FREE DEBT

- With $\beta_T = 0.9$: $r_T = r^f + 0.9 \times RP = 1.5\% + 2.7\% = 4.2\%$.
- Price-dividend ratio on taxes: $1/(r_T - x) = 1/(0.042 - 0.02) = 45.5$.
- Average tax/GDP: $\tau = 22\%$ (lowered from 25%). Expected primary deficit: 0.5% of GDP.

Assets		Liabilities	
$PV(T)$	$\$2.2 \times 45.5 = \$100$	$PV(G)$	$\$2.25 \times 40 = \$90$
		D	$\$10$
Total	$100	Total	$100

- Note: taxes use pd = 45.5 ($\beta_T = 0.9$), spending still uses pd = 40 ($\beta_G = 1$).
- The government can on average run a deficit of 0.5% of GDP while keeping debt risk-free.

THE HIDDEN COST OF RISK-FREE DEBT: SURPLUSES IN RECESSIONS

- How can debt/GDP = 100% while the expected surplus is –0.5% of GDP?
- The government provides insurance to bondholders: it delivers positive surpluses when GDP growth is lower than average.
- Bondholders pay an insurance premium of 0.5% of GDP to receive larger payments in high-marginal-utility states.
- But providing this insurance is costly: it requires the government to raise taxes or cut spending in recessions.
- Less room for output stabilization – the government cannot simultaneously insure taxpayers and bondholders.

HIGHER DEBT REQUIRES AN EVEN SAFER TAX PROCESS

- At $d = 200\%$: $\beta_T = \frac{90}{90+20} \cdot 1 = 0.82$.
- Tax revenue must become even more countercyclical.
- Average tax/GDP falls from 22% to $\approx$ 21.5%... but the present value of the tax claim rises from \$100 to \$110 because the lower β_T raises the price-dividend ratio.
- Taxpayers owe more in present value. They provide more valuable insurance to bondholders.
- The constraint on fiscal policy tightens as debt grows.

CONVENIENCE YIELDS

GOVERNMENT DEBT PROVIDES LIQUIDITY AND SAFETY BEYOND ITS CASH FLOWS

- Sometimes government debt is more valuable than the PDV of its cash flows.
- It provides liquidity and safety to investors – similar to holding cash despite its negative real return.
- The convenience yield λ_t: the extra return investors forgo for the privilege of holding Treasuries.

$$R_t^{\text{Treas}} = R_t^f - \lambda_t.$$

- Treasury yields are below the true risk-free rate by the convenience yield.
- Important: this does not change the riskiness of surpluses (still discounted at $r^f + RP$).
- Instead, it adds a new item to the Treasury balance sheet: seignorage revenue $K_t = (1 - e^{-\lambda_t})D_t$.
- Each period, the government sells new bonds at a price above fundamental value (investors accept below-r^f yields for the liquidity services). This price premium is revenue.

CONVENIENCE YIELDS AVERAGE 73 BPS FOR US TREASURIES

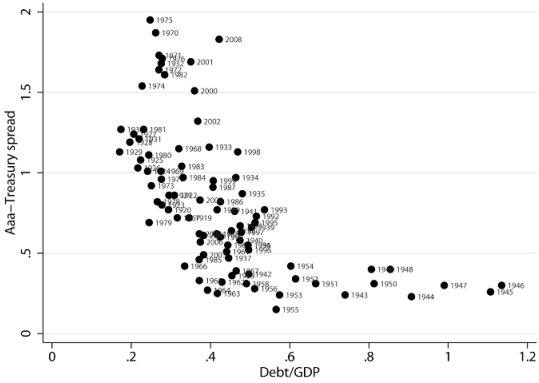


FIG. 1.—Corporate bond spread and government debt. The figure plots the Aaa-Treasury corporate bond spread (y axis) against the debt-to-GDP ratio (x axis) on the basis of annual observations from 1919 to 2008. The corporate bond spread is the difference between the percentage yield on Moody's Aaa long-maturity bond index and the percentage yield on long-maturity Treasury bonds.

Source: Krishnamurthy & Vissing-Jorgensen (2012)

CONVENIENCE YIELDS MODIFY THE IGBC WITH SEIGNORAGE REVENUE

- The modified IGBC:

$$B_t = \mathbb{E}_t \sum_{j=1}^{\infty} M_{t,t+j} \left(T_{t+j} - G_{t+j} \right) + \mathbb{E}_t \sum_{j=1}^{\infty} M_{t,t+j} B_{t+j} \left(1 - e^{-\delta_{t+j}} \right).$$

- The new term: seignorage revenue $K_{t+j} = B_{t+j}(1 - e^{-\delta_{t+j}})$.
- The government earns revenue by selling bonds at a premium over fundamentals.
- The cost of funding becomes:

$$r_D = \frac{PV(T)}{D} r_T - \frac{PV(G)}{D} r_G + \frac{PV(K)}{D} r_K.$$

THE TRUE RISK-FREE RATE IS HIGHER THAN TREASURY YIELDS

- Treasury yields are not the true risk-free rate. The true r^f is the yield an investor would earn on a riskless asset with no liquidity or safety services.
- With a convenience yield of 73 bps: $r^f = 1.5\% + 0.73\% = 2.23\%$.
- This raises the discount rate for all risky claims: $r_Y = r^f + RP = 2.23\% + 3\% = 5.23\%$.
- The price-dividend ratio on GDP falls: $1/(5.23\% - 2\%) = 30.96$ (vs. 40 before).
- PDV of surpluses falls from 100% to $\approx$ 77% of GDP.

SEIGNORAGE OFFSETS THE HIGHER DISCOUNT RATE – ALMOST EXACTLY

- Seignorage revenue adds fiscal capacity: $\approx$ 23% of GDP (if $\beta_K = 1$).
- Net fiscal capacity: 77% + 23% = 100% of GDP – same as before.
- Convenience yields are not a free lunch: you gain seignorage revenue, but the true r^f is higher than Treasury yields, which raises the discount rate for surpluses. The two effects roughly cancel.

COUNTER-CYCLICAL CONVENIENCE YIELDS HELP MORE

- In the short run, convenience yields can be counter-cyclical: flight to safety in crises.
- This means $\beta_K < 1$: seignorage revenue is less risky than GDP.
- Lower β_K increases the present value of seignorage without affecting the risk-free rate.
- With $\beta_K = 0.584$: fiscal capacity rises to 114% of GDP.
- But the gain is modest: only 14 percentage points above the no-convenience-yield case.
- Convenience yields matter, but they do not resolve the fiscal capacity puzzle.

QUANTIFYING US FISCAL CAPACITY

MEASURING FISCAL CAPACITY REQUIRES TWO INPUTS: CASH FLOWS AND DISCOUNT RATES

- Jiang et al. (2022, *Brookings Papers*) implement the DCF approach using CBO projections.
- Cash flows: CBO projects federal tax revenue and non-interest spending for 2022–2052. Average primary surplus: −3.2% of GDP.
- Discount rate: for each horizon h, set

$$r^{s,y}(h) = y^f(h) + \underbrace{rp^y}_{\text{GDP risk premium}},$$

  where $y^f(h)$ is the nominal zero-coupon Treasury yield at maturity h.
- This is the expected return on a claim to GDP. Since surpluses are riskier than GDP ($\beta_S > \beta_Y$), the true discount rate for surpluses is higher. Using the GDP rate therefore produces an upper bound on fiscal capacity.

THE GDP RISK PREMIUM COMES FROM UNLEVERING THE EQUITY RISK PREMIUM

- Expected real return on US equities (end of 2021): $\approx 3.6\%$.
- Real risk-free rate: -1.5%. Equity risk premium: $\approx 5.1\%$.
- Unlever using the corporate debt/equity ratio of 1:2 (equity/assets = 2/3):

$$rp^y = \frac{2}{3} \times 5.1\% = 3.4\%.$$

- Subtract the long-bond term premium ($\approx 0.8\%$) to get the GDP risk premium over long-term bonds: $rp^y \approx 2.6\%$.
- This is conservative: the historical average excess stock return is $\approx 8\%$ (1947–2021). Applying the same unlevering gives $rp^y \approx \frac{2}{3} \times 8\% - 0.8\% \approx 4.5\%$.

CBO PROJECTS PERSISTENT DEFICITS; THE IGBC IMPLIES UNPRECEDENTED SURPLUSES

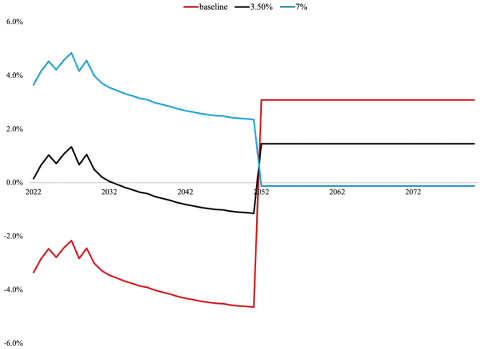


Figure 1: CBO Baseline Projections of Primary Surplus (baseline in red, baseline +3.5% in black and baseline +7% in blue) between 2022 and 2051. Starting in 2052, steady-state surplus is chosen to match debt/GDP ratio in 2052.

Source: Jiang et al. (2022, BPEA), Figure 1.

FISCAL CAPACITY DECOMPOSES INTO NEAR-TERM DEFICITS AND FAR-FUTURE SURPLUSES

- Split the PDV of all future surpluses at the CBO projection horizon:

$$\underbrace{PV_{2021}\left(\{S\}_{2022}^{\infty}\right)}_{\text{fiscal capacity}} = \underbrace{PV_{2021}\left(\{S\}_{2022}^{2052}\right)}_{\text{CBO projections}} + \underbrace{PV_{2021}\left(D_{2052}\right)}_{\text{tail value}}.$$

- CBO projects debt/GDP = 185% in 2052. To back this, the US needs a permanent surplus of S/Y after 2052:

$$\frac{D_{2052}}{Y_{2052}} = \frac{S}{Y} \times pd^{y} \quad \Rightarrow \quad \frac{S}{Y} = \frac{185\%}{85.8} = 2.16\%.$$

- This requires a fiscal correction of ≈ 5.4 pp relative to the CBO average of −3.2%.

WHAT IS DURATION?

- We computed fiscal capacity at a given discount rate. But how sensitive is this number to changes in rates?
- Duration measures the weighted average time until a cash flow stream pays out.
- For a stream of payments $CF_1, CF_2, \ldots,$ Macaulay duration is:

$$\text{Duration} = \frac{\sum_{j=1}^{\infty} j \times PV(CF_j)}{\sum_{j=1}^{\infty} PV(CF_j)}.$$

- A 5-year zero-coupon bond has duration = 5 years.
- Duration also measures interest rate sensitivity: a 1 pp rise in the discount rate reduces the present value by approximately Duration $\times$ 1%.
- Example: a claim with duration 100 years loses $\approx$ 100% of its value for a 1 pp rate increase.

THE DURATION MISMATCH EXPOSES THE GOVERNMENT TO INTEREST RATE RISK

- The surplus claim is back-loaded: deficits until 2052, surpluses after.
- Duration of the surplus claim: ≈ 283 years.
- Duration of the outstanding Treasury portfolio: ≈ 5 years.
- A 1 pp increase in rates barely changes the value of liabilities, but dramatically reduces the value of the surplus claim.
- Between Dec 2021 and May 2022, rates rose 130–175 bps. Required post-2052 surplus jumps from 2.16% to 6.24% of GDP.
- To hedge: lengthen debt maturity (so rate rises also reduce the value of liabilities) or front-load surpluses (shorten the duration of the asset side). But matching 283 years of duration is not realistic with any debt structure.

THE SURPLUS CLAIM'S DURATION IS 283 YEARS

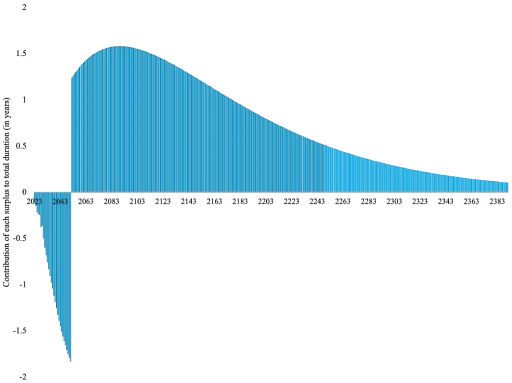


Figure 2: Duration Composition in Baseline Scenario. Contribution of each payment $\frac{k \times PV(S_{2021+k})}{\sum_{j=0} PV(S_{2021+j})}$ to total duration of $\frac{\sum_{k=0} k \times PV(S_{2021+k})}{\sum_{j=1} PV(S_{2021+j})}$ in CBO Baseline Projection. Units in years. The duration is the sum of the plotted contributions.

Duration decomposition of the surplus claim.

TWO APPROACHES TO MEASURING THE DISCOUNT RATE

- Approach 1 (upper bound – BPEA paper): discount all fiscal cash flows at the GDP rate $r^{s,y}(h) = y^f(h) + rp^y$.
  - Overvalues the tax claim (uses too low a rate) and undervalues the spending claim (too high a rate).
  - Produces an upper bound on fiscal capacity.
- Approach 2 (full SDF – *Econometrica* paper): specify an exponentially affine SDF $M^{\$}_{t,t+j}$ whose shocks are priced aggregate risk factors.
  - Calibrate market prices of risk to match the term structure of nominal and real bond yields and the stock market price-dividend ratio (1947–2020).
  - The SDF produces time-varying, maturity-specific discount rates for taxes and spending separately.
  - Not an upper bound but the actual PDV. Result: fiscal capacity averages −229% of GDP.

SENSITIVITY: THE GDP RISK PREMIUM IS THE KEY PARAMETER

- All results hinge on the GDP risk premium rp^y.
- Baseline: $rp^y = 2.6\%$, fiscal capacity = \$12.4T. Outstanding debt = \$22.3T. Gap of \$10 trillion.
- What rp^y would close the gap? Reverse-engineering: $rp^y = 1.37\%$. But at that premium (eq. 15 in Jiang et al.):

$$r^{s,y} - g = y^f + rp^y - g = 2.07\% + 1.37\% - 3.50\% < 0.$$

  The discount rate falls below the growth rate, the TVC fails, and fiscal capacity is infinite.
- This is the $r < g$ argument from last time (Blanchard 2019) – it implicitly sets $rp^y = 0$.
- Recall: Abel et al. (1989) showed the US is dynamically efficient despite $r^f < g$. The risk premium is not zero.
- With $rp^y = 0$, an unlevered firm growing at the GDP rate would have infinite valuation – inconsistent with observed stock prices.

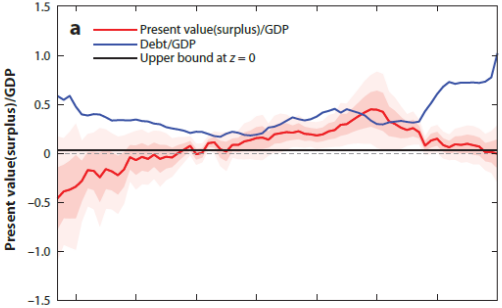
CBO PROJECTIONS IMPLY FISCAL CAPACITY $8 TRILLION BELOW OUTSTANDING DEBT

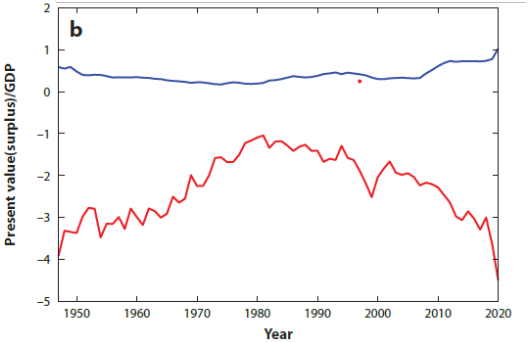
- Jiang et al. (2023, Table 4) present the Treasury balance sheet using CBO projections for 2022–2052.

Assets		Liabilities	
$PV(T_{2021}^{2051})$	$124.2	$PV(G_{2021}^{2051})$	$145.4
$PV(S_{2051}^{\infty})$	$35.2	D	$14.1
Total	$159.5	Total	$159.5

- Outstanding debt at end of 2021: $22.3 trillion. Gap: $8.2 trillion.

VAR ESTIMATES CONFIRM: DEBT EXCEEDS FISCAL CAPACITY SINCE THE 2000S





Panel (a): upper bound on fiscal capacity. Panel (b): SDF-based estimate.

THE US GOVERNMENT BOND VALUATION PUZZLE

- The gap between the market value of debt and estimated fiscal capacity is the government bond valuation puzzle.
- Panel (a): the upper bound on fiscal capacity (using the GDP discount rate for both taxes and spending) has been below debt/GDP since the early 2000s.
- Panel (b): the SDF-based fiscal capacity measure averages –229% of GDP – far below debt/GDP of 39% on average.
- The valuation gap averages 268% of GDP and has widened dramatically since the GFC and COVID.

CANDIDATE EXPLANATIONS FOR THE PUZZLE

- Convenience yields? Too small to close the gap (~ 14 pp).
- Bubble in Treasuries? Hard to sustain when aggregate risk is properly priced ($r^f + RP > x$).
- Global safe asset supplier? US has unique role; investors coordinate on US debt, reducing rollover risk. UK lost this status after WWII.
- Mispricing / investor optimism? Debt-implied surpluses have persistently exceeded realized surpluses since the late 1990s.
- Fiscal correction? Would require unprecedented surpluses with high probability every year.
- QE and financial repression? Central banks bought large fractions of new issuance.

THE US IS AN OUTLIER – DEVELOPING COUNTRIES FACE MARKET DISCIPLINE

- In developing countries: bad fiscal news $\Rightarrow$ bond price declines. Not so in the US.
- Developing countries have procyclical surpluses (Talvi & Vegh, 2005; Mendoza & Ostry, 2008), creating substantial fiscal risk.
- Developing countries have higher average surpluses and lower debt/GDP.
- UK comparison: before WWII, UK was the global safe asset supplier. Fiscal capacity = 68% of GDP, debt/GDP = 87%. After WWII: fiscal capacity = 82%, debt/GDP = 53%.
- After ceding safe-asset status to the US, UK lost convenience yields and had to cut debt.

MONETARY POLICY AND FISCAL CAPACITY

“THE FED CONTROLS RATES, SO DEBT SUSTAINABILITY IS NOT AN ISSUE”

- A common argument: the central bank sets interest rates, so it can always keep debt service affordable.
- This argument conflates two different objects:
  1. The nominal short rate – which the Fed does control.
  2. The risk premium – which the Fed does not control.
- Even if Treasury yields are low, the true fiscal cost includes the GDP risk premium because surpluses are tied to GDP.
- The Fed can lower the rate bondholders receive, but it cannot lower the rate at which the government’s surplus stream should be discounted.

CONSOLIDATING THE FED AND TREASURY DOES NOT CHANGE FISCAL CAPACITY

- Jiang et al. (2022): “Consolidation would not change the amount of government liabilities held by the private sector. It would merely imply a shortening of the maturity structure.”
- QE: the Fed buys long-term bonds and issues short-term reserves.
- Effect on the private sector: long-duration Treasuries are replaced by short-duration reserves.
- The PDV of future surpluses is unchanged. QE does not create new fiscal capacity.
- Worse: shortening the duration of public-held debt exacerbates the maturity mismatch (surplus duration $\approx$ 283 years vs. reserves $\approx$ 0).

QE HELPS DURING CRISES BUT CANNOT ELIMINATE AGGREGATE RISK

- Elenev et al. (2021): QE can help during a crisis at the ZLB.
  - Provides high-quality collateral to intermediaries, stimulates aggregate demand.
  - Large fiscal multiplier at ZLB $\Rightarrow$ higher output $\Rightarrow$ lower debt/GDP.
  - Lowers long-term rates, reducing debt service costs.
- But: the central bank's losses from QE are ultimately borne by taxpayers (reduced remittances to Treasury).
- The consolidated government balance sheet is unchanged.
- Monetary policy can redistribute risk – but it cannot create fiscal capacity out of nothing.

FINANCIAL REPRESSION IS A HIDDEN TAX ON BONDHOLDERS

- After WWII, the US pegged nominal interest rates below inflation (Acalin & Ball, 2022).
- Combined with capital controls forcing institutions to hold government debt.
- This is a hidden tax on bondholders: they earn negative real returns, effectively transferring wealth to the government.
- Recent parallels: ECB purchased more bonds than member states issued during the pandemic; Fed bought essentially all new long-term issuance.
- If bond prices are artificially high due to central bank purchases, the “puzzle” partly reflects policy distortions.
- But repression has limits: it erodes the credibility of the safe asset status and drives away voluntary investors.

THE CORRECT FISCAL COST METRIC ACCOUNTS FOR THE GDP RISK PREMIUM

- The fiscal cost of debt is not $r^f - x$ (which can be negative).
- The correct metric is $r^f + RP - x$: the risk-free rate plus the GDP risk premium minus growth.
- Using $r^f + RP - x = 2.5\%$ instead of $r^f - x = -0.5\%$ transforms fiscal capacity from ∞ to 100% of GDP.
- The government cannot make aggregate risk disappear through financial engineering.
- Bohn's fiscal reaction function ($\rho > 0$) is necessary but not sufficient: even with $\rho > 0$, the risk premium determines the true fiscal burden.
- The trade-off between insuring taxpayers and insuring bondholders is fundamental.